



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 – 15



**KANEKO Single-Pass Rice Mill (C)
Model: Mark III**

KANEKO MARK III SINGLE-PASS RICE MILL (C)

Test Report No.	:	2013 - 15
Date of Test	:	February 8, 2013
Test Valid Until	:	February 8, 2018
Brand	:	KANEKO
Model	:	MARK III
Type	:	Rubber roll huller - friction whitener, single-pass rice mill
Manufacturer	:	SEACOM MANUFACTURING, INC. 721 Aurora Blvd., New Manila Quezon City
Test Requested By	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The KANEKO MARK III Single-Pass Rice Mill was tested using an unknown rice variety with average moisture content of 13.6%. This was the third test for the same model requested by the same manufacturer. The first test was conducted on April 26, 2005 with Test Report No. 2005-15 while the second test was conducted on February 18, 2011 with Test Report No 2011-11.

It had an average Input Capacity of 842.2 kg/h palay and an average Output Capacity of 562.0 kg/h milled rice. Test results showed that the rice mill had an average Milling Recovery of 66.7%.

Laboratory analysis of milled rice samples showed that based on milled rice, Head Rice Recovery was 60.0% with 27.3 % Broken Grains, 12.7% Brewer's Rice and no Unhulled Grains. Milling Degree analysis showed that the milled rice output fell within the well-milled rice classification.

The average Fuel Consumption of the MITSUBISHI 4DR5 diesel engine was 1.63 L/h.

During the tests two persons were needed to operate the machine.

No machine breakdown or failure of components occurred during the tests.

METHOD AND SCOPE OF TEST

The rice mill was tested for verification of its performance. The test was based on the Philippine Agricultural Engineering Standard "PAES 207:2000 Agricultural Machinery – Rice Mill – Methods of Test". The test was conducted to verify and assess the following:

1. Capacities
 - a. Input capacity
 - b. Output capacity
2. Total milling recovery
3. Quality of milled rice output
 - a. Head rice recovery
 - b. Broken rice
 - c. Milling degree
 - d. Whiteness index

PLACE AND DATE OF TEST

The rice mill was tested at the plant site of SEACOM, INC. in Sucat, Parañaque City on February 8, 2013.

OPERATOR'S MANUAL

The operator's manual provided with the machine contains the following: Features; Specifications; Assembly; Installation; Repair and Maintenance; Cautions; and, Parts List.

DESCRIPTION OF THE MACHINE

The KANEKO MARK III Rice Mill (**Figure 1**) was a one-pass rice mill, consisted of the following components: hopper, hulling unit whitening unit, aspirating units for rice hull and rice bran; and, the prime mover.

The hopper held the paddy for milling. The rate of feeding was controlled by a sliding gate (**Figure 2**) and a paddy feed adjuster (valve adjuster) together with the feed roll installed at the bottom of the hopper. The hulling assembly consisted of a pair of 102 mm x 222 mm (W x D) drum-type rubber roll hullers (**Figure 3**) rotating in opposite directions at different speeds. Clearance between the hullers was adjusted through the rubber roll adjuster. Rice hulls were discharged through the hull aspirator.

Brown rice and some unhulled paddy discharged from the hulling unit entered the whitening chamber for bran removal. The whitening unit (**Figure 4**) consisted of a friction-type resistance roll and one pair of hexagonal-shaped screen. The retention time of the grain inside the whitening chamber was controlled by the adjustable sliding counter weight at the grain outlet spout where milled rice was discharged and flowed through a chute. Bran was discharged through the hexagonal screen with an air blast provided by a centrifugal fan (**Figure 5**). The fan also helped clean and cool the rice during the whitening process.

The machine was powered by a MITSUBISHI 4DR5 diesel engine (**Figure 6**) which drove directly the whitener main shaft through a quintuple B-section pulley and V-belts. The step down pulley at the whitener main shaft drives the two hullers, belt tightener and the air blast fan through a double grooved A-section pulleys and hexagonal belts. The power transmission system of the rice mill is shown in **Figure 7**.

Detailed specifications of the rice mill are shown in **Table 1**.



Figure 1. KANEKO Single-Pass Rice Mill
Model: Mark III

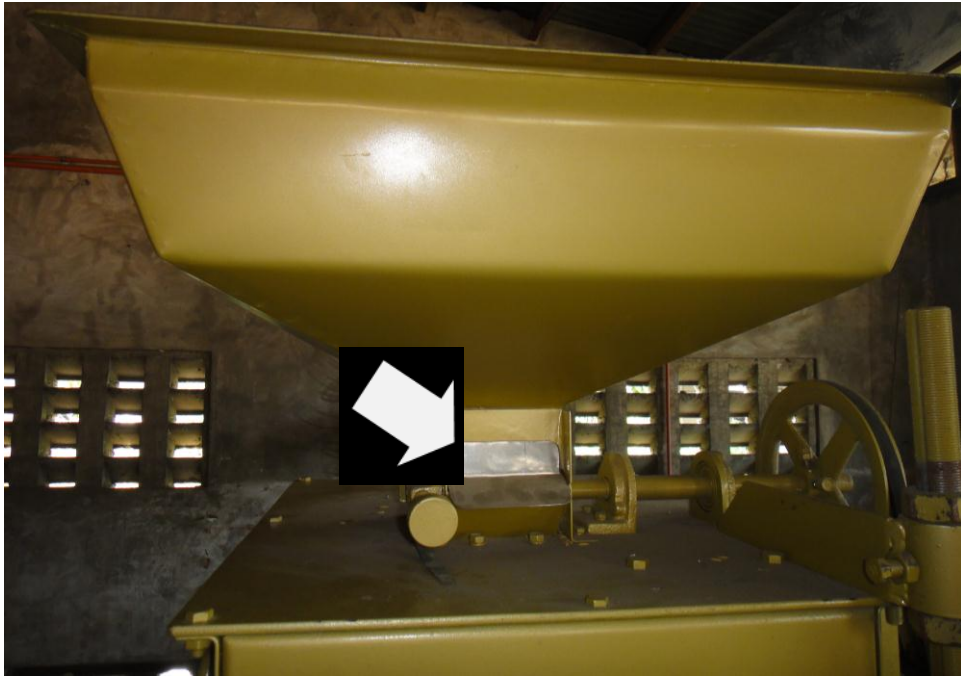


Figure 2. Sliding gate



Figure 3. Hulling unit

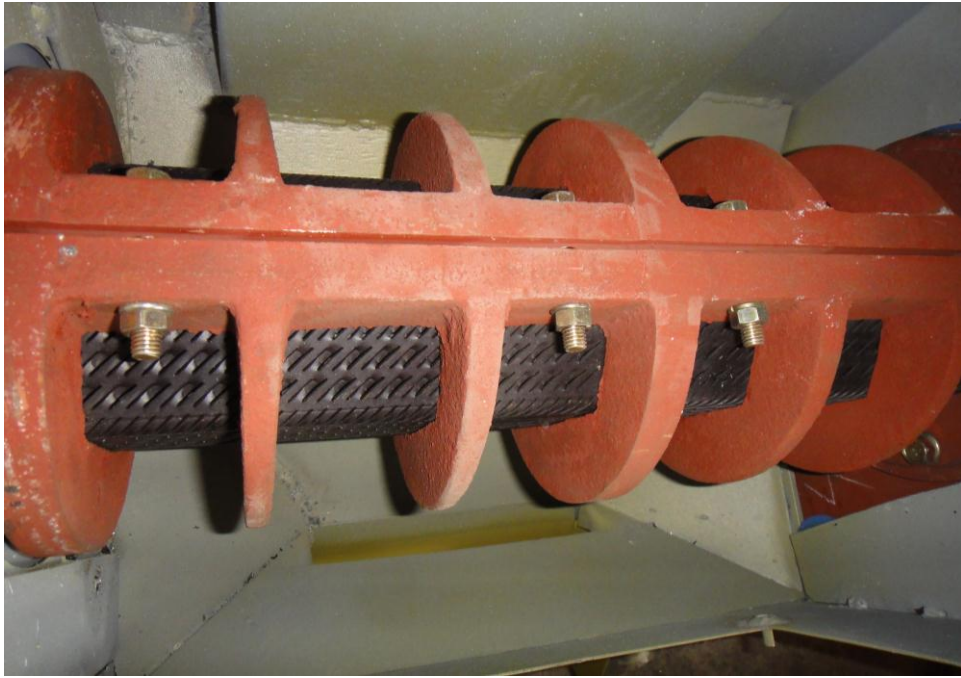


Figure 4. Whitening unit

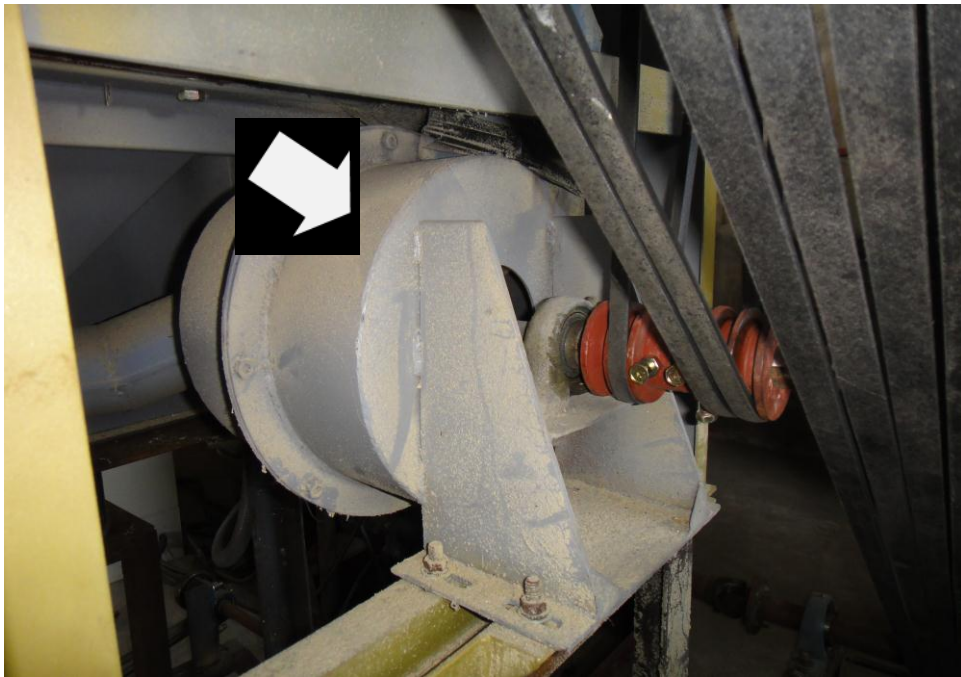


Figure 5. Air blast fan

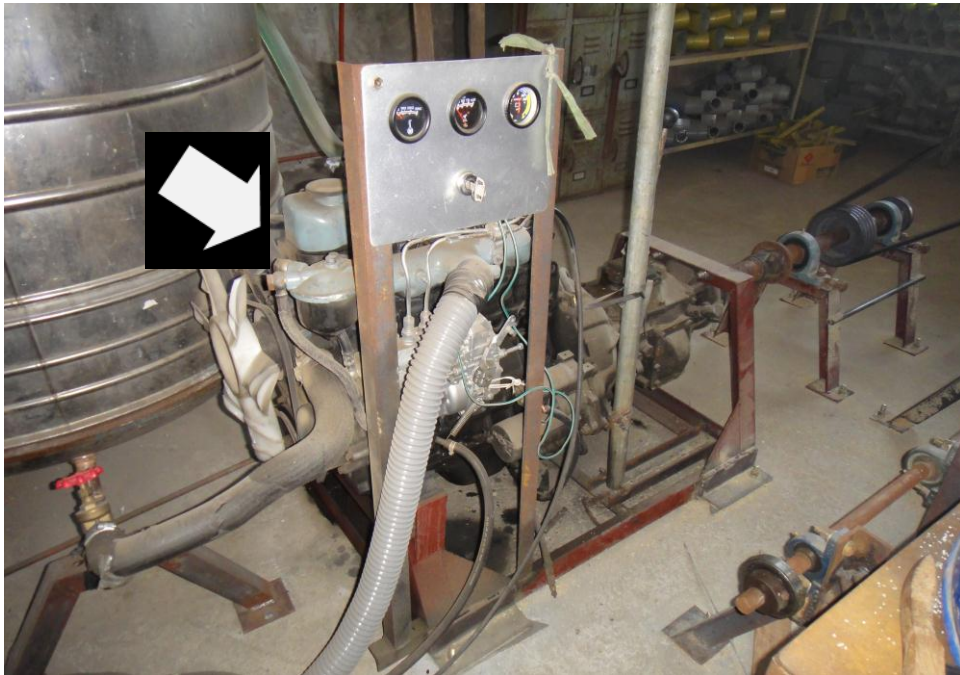


Figure 6. Prime mover



Figure 7. Belt-pulley power transmission system

RESULTS AND DISCUSSION

Test Material

“Unknown” rice variety was used during the test. General information on the test material is shown in **Table 2**.

Input Capacity

The Input Capacity of a rice mill refers to the amount of paddy fed to the rice mill per unit time. The machine had an average Input Capacity of 842.2 kg/h.

Output Capacity

The Output Capacity of a rice mill refers to the amount of milled rice output of the rice mill per unit time. The machine had an average Output Capacity of 562.0 kg/h.

Milling Performance (Please refer to **Table 3**)

The Milling Performance of a rice mill is expressed in terms of its Milling Recovery, Head Rice Recovery and Milling Degree. (See Annex 1 for formula/definition).

Milling Recovery and Milling Degree. The Milling Recovery of the rice mill was 66.7% with milling degree classified as well-milled.

Head Rice Recovery. Head rice was separated from broken grains using laboratory indented trays. Results showed that the Head Rice Recovery was 60.0% based on milled rice and 40.0% based on rough rice.

Fuel Consumption

The average Fuel Consumption of the MITSUBISHI 4DR5 diesel engine was 1.63 L/h.

Labor Requirement

During the test, two persons (**Figure 8**) were needed to operate the rice mill: one for feeding and making the necessary adjustments on the machine and one for collecting of bran and bagging of milled rice.

Observation

No machine breakdown or failure of components occurred during the tests.



Figure 8. Operators of the rice mill

Table 2. Laboratory analysis of paddy samples

ITEMS	DATA
Variety	“Unknown”
Source	San Mateo, Rizal
<u>Paddy Analysis</u>	
Grain Size, brown rice (mm)	
Length	7.75
Width	2.13
Thickness	1.66
Grain Type	long and slender
Paddy Bulk Density (kg/m ³)	532.7
Moisture Content, wet basis (%)	13.6
Purity (%)	99.28
Cracked Grains (%)	3.33
Immature Grains (%)	0.72
Chalky Grains (%)	7.32
Yellow and Fermented Grains (%)	4.44
Red Rice (%)	Nil
<u>Laboratory Milling Test^{1/}</u>	
Coefficient of Hulling	0.94
Coefficient of Wholeness	0.88
Hulling Efficiency (%)	82.7
Total Milling Recovery (%)	66.9

^{1/} Satake Laboratory Test Mill

Table 2. (cont'n.)

ITEMS	DATA
Head Rice Recovery (%)	
Based on Milled Rice	80.2
Based on Rough Rice	53.6
Broken Rice (%)	
Based on Milled Rice	17.7
Based on Rough Rice	11.8
Brewer's Rice (%)	
Based on Milled Rice	2.1
Based on Rough Rice	1.4
Whiteness Index	39.8
Milling Degree *	Well-milled

*Based on alcohol-alkali bran staining method for determining milling degree of rice

Table 2. Machine performance test results

	DATA
A. Milling Performance	
1. Total Weight of Input Paddy (kg)	463.2
2. Total Weight of Milled Rice (kg)	309.1
3. Total Operating Time (h)	0.55
4. Input Capacity (kg/h)	842.2
5. Output Capacity (kg/h)	562.0
6. Total Milling Recovery (%)	66.7
7. Coefficient of Hulling	0.89
8. Coefficient of Wholeness	0.92
9. Hulling Efficiency (%)	81.9
B. Milled Rice Analysis	
1. Head Rice Recovery (%)	
Based on Milled Rice	60.0
Based on Rough Rice	40.0
2. Broken Grains (%)	
Based on Milled Rice	27.3
3. Brewer's Rice (%)	
Based on Milled Rice	12.7
4. Impurities (%)	
Based on Milled Rice	0

Table 2. (cont'n.)

		DATA
5.	Unhulled Grains (%)	
	Based on Milled Rice	0
6.	Whiteness Index	39.7
7.	Milling Degree *	Well-milled
C.	Analysis of Bran from the Whitening Unit	
	a) Bran (%)	97.0
	b) Broken and Brewer's Rice (%)	3.0
D.	Speed of Components (average rpm)	
1.	Prime Mover	
	Without Load	1409
	With Load	1369
2.	Feed Roll	
	Without Load	490
	With Load	470
3.	Rubber Roll #1 (adjustable)	
	Without Load	1276
	With Load	1219
4.	Rubber Roll #2 (fixed)	
	Without Load	963
	With Load	930

* Based on alcohol-alkali bran staining method for determining milling degree of rice

Table 2. (cont'n.)

		DATA
5.	Hull Aspirator	
	Without Load	1153
	With Load	1120
6.	Whitener Main Drive	
	Without Load	779
	With Load	758
7.	Air Blast Fan	
	Without Load	3783
	With Load	3672
E.	Noise Level [db(A)]	
1.	Feeder	
	Without Load	81.2
	With Load	87.1
2.	Bagger	
	Without Load	84.6
	With Load	89.6
F.	Prime mover (MITSUBISHI 4DR5 Diesel Engine)	
	Fuel Consumption (L/h)	1.63

ANNEX 1. FORMULA USED IN THE COMPUTATION/DEFINITION OF TERMS

$$\text{a) Input Capacity (kg/h)} = \frac{\text{Weight of clean paddy (kg)}}{\text{Input time (h)}}$$

$$\text{b) Output Capacity (kg/hr)} = \frac{\text{Weight of milled rice (kg)}}{\text{Output time (h)}}$$

$$\text{c) Milling Recovery (\%)} = \frac{\text{Weight of milled rice (kg)}}{\text{Weight of clean paddy (kg)}} \times 100$$

$$\text{d) Coefficient of Hulling, } H_c$$

$$H_c = 1 - \frac{\text{wt. of unhulled paddy}}{\text{wt. of sample (mixture of brown rice and unhulled paddy)}}$$

$$\text{e) Coefficient of Wholeness, } W_c$$

$$W_c = \frac{W_s}{t_s}$$

where : W_s = the weight of the whole brown rice
 t_s = the weight of the total hulled samples

$$\text{f) Hulling Efficiency, } H_e (\%)$$

$$H_e = (\text{Coefficient of Hulling}) \times (\text{Coefficient of Wholeness}) \times 100$$

$$\text{g) Head Rice Recovery (\%)*} = \frac{\text{wt. of head rice (g)}}{\text{wt. of milled rice (g)}} \times 100$$

$$\text{h) Broken Rice (\%)*} = \frac{\text{Weight of broken rice (g)}}{\text{Weight of milled rice (g)}} \times 100$$

$$\text{i) Brewer's Rice (\%)*} = \frac{\text{Weight of brewer's rice (g)}}{\text{Weight of milled rice (g)}} \times 100$$

j) Milling Degree - The extent or degree by which different bran layer that covers the brown rice is removed as a result of whitening.

* Based on a 100-gram sample of milled rice

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N.B.

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